

# The Ordinal Summit

*Where the Tower Snaps to  $c_1$*

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This companion explains the ideas in plain language with diagrams and examples. It is not the formal document — every claim here restates a result already proved in ZP-L Incomputability Convergence. Consult that document for the authoritative mathematics.

## What Is ZP-L Doing?

ZP-K carries the computational face of the initial machine state  $c_0$  as a KleeneStructure requirement (botCode\_is\_quine): a program whose behavior is determined entirely by its own index, whose existence the second recursion theorem guarantees. Reading  $c_0$  as the Kleene quine is that commitment, not a theorem. ZP-L picks up the thread from a different direction: the ordinal direction. If a map  $\phi$  assigns machine phases to ordinals — sending  $c_0$  to everything below a certain point and  $c_1$  above it — what is that threshold ordinal? ZP-L establishes the answer:  $\varepsilon_0$ , the first fixed point of the map  $\alpha \mapsto \omega^\alpha$ .

This is not an arbitrary choice. ZP-L proves that  $\varepsilon_0$  is the minimal threshold — no monotone, tower-aligned map can snap before  $\varepsilon_0$ , and the canonical map snaps exactly there. ZP-L also establishes the connection to ZP-B: as the tower stages approach  $\varepsilon_0$  in ordinal order, their 2-adic encodings converge to  $0 = \perp$  in  $\mathbb{Z}_2$ . The same sequence witnesses both convergences simultaneously.

### Real-world analogy — the unreachable horizon

Imagine a path where each step is longer than the last: you walk 1 mile, then 2, then 4, then 8, ... You are always getting further from the start, but you are also always somewhere on the path — never at the horizon. The horizon ( $\varepsilon_0$ ) is the limit of all those steps combined. It exists as a mathematical object, but no finite number of steps reaches it.

The ordinal tower works the same way. The snap that ZP-E proved occurs at  $\varepsilon_0$  is not a step you can take from inside the tower. It is the point where the tower stops and something new begins.

## What Is $\varepsilon_0$ ?

$\varepsilon_0$  (epsilon-zero) is the smallest ordinal  $\alpha$  satisfying the equation  $\omega^\alpha = \alpha$ . In other words, it is the first fixed point of the map that sends any ordinal  $\alpha$  to  $\omega^\alpha$  (omega raised to the power  $\alpha$ ). This is not a coincidence or a definition by fiat — it is a property  $\varepsilon_0$  satisfies and no smaller ordinal does.

The tower  $0, 1, \omega, \omega^\omega, \omega^{\omega^\omega}, \dots$  climbs through the ordinals, each stage being  $\omega$  raised to the previous stage. Every stage is strictly below  $\varepsilon_0$ , and  $\varepsilon_0$  is their limit — the supremum of the whole sequence. It is never reached from below; it is the boundary above all stages.



*Each tower stage is strictly below  $\varepsilon_0$  and maps to  $c_0$ . The snap to  $c_1$  occurs exactly at  $\varepsilon_0$ .*

The tower stages  $0, 1, \omega, \omega^\omega, \dots$  all lie strictly below  $\varepsilon_0$  and map to  $c_0$ . The snap transition to  $c_1$  occurs at the threshold  $\varepsilon_0$  itself. The arrow marks the snap. ZP-L §VII: `epsilon_zero_snap_canonical`.

Remember:  $\varepsilon_0$  is not "infinity." It is a specific countable ordinal — the first one satisfying  $\omega^\alpha = \alpha$ . Ordinals larger than  $\varepsilon_0$  exist ( $\varepsilon_0 + 1, \varepsilon_1$ , and many more). ZP-L is not claiming that  $\varepsilon_0$  is the largest ordinal or that the framework breaks down above it. It is the minimal snap threshold for monotone, tower-aligned maps — and that minimality is exactly what is proved.

## Why $\varepsilon_0$ Is the Exact Threshold

Two facts together pin  $\varepsilon_0$  as the minimal snap threshold. They work like a lower bound and an upper bound that meet at the same point.

**Lower bound (snap cannot happen before  $\varepsilon_0$ ):** The fundamental sequence is cofinal in  $\varepsilon_0$ . This means: for any ordinal  $\alpha$  below  $\varepsilon_0$ , some tower stage eventually exceeds  $\alpha$ . A monotone map that assigns  $c_0$  to every tower stage must therefore assign  $c_0$  to  $\alpha$  too (it cannot increase between a stage above  $\alpha$  and  $\alpha$  itself). So no ordinal strictly below  $\varepsilon_0$  can be a snap point for such a map.

**Upper bound (snap does happen at  $\varepsilon_0$ ):**  $\varepsilon_0$  is itself a fixed point of  $\omega^\cdot$  ( $\omega^\varepsilon_0 = \varepsilon_0$ ). A map that assigns  $c_1$  to all fixed points of  $\omega^\cdot$  must assign  $c_1$  to  $\varepsilon_0$ . Together with the lower bound,  $\varepsilon_0$  is the first place where  $c_1$  can and must appear.

The canonical witness map  $\phi(\alpha) = c_0$  if  $\alpha < \varepsilon_0$ , else  $c_1$  satisfies all five conditions at once — monotonicity, tower-alignment, fixed-point-respecting, snapping at  $\varepsilon_0$ , and minimality — with no free hypotheses. This is the central result of ZP-L.

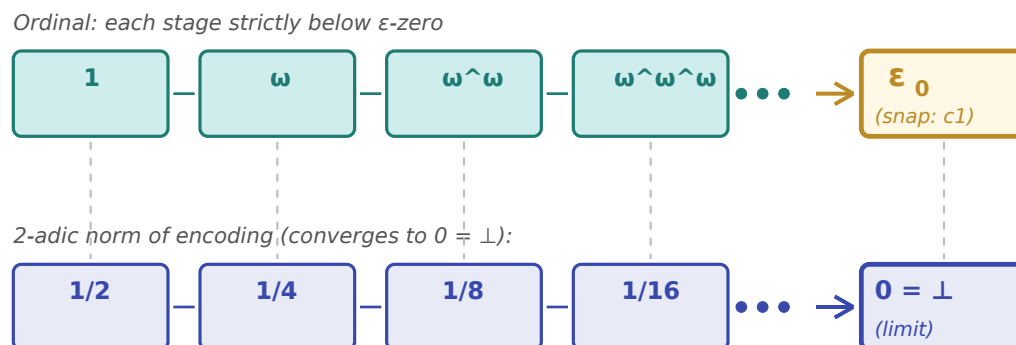
"Minimal" does not mean "unique." ZP-L proves that no snap can occur strictly before  $\varepsilon_0$  (under the stated conditions), not that  $\varepsilon_0$  is the only possible snap point. Maps that also assign  $c_1$  to ordinals above  $\varepsilon_0$  are not ruled out. The framework fixes  $\varepsilon_0$  as the threshold by the canonical witness, not by uniqueness of all satisfying maps.

## The Two-Adic Connection

ZP-B established that  $0 = \perp$  in the 2-adic integers  $\mathbb{Z}_2$  is the valiative sink: the 2-adic valuation of 0 is infinite, which means 0 sits at the bottom of the 2-adic metric. ZP-L connects this to the ordinal tower via a map,

`cnfToZp2`, from ordinals below  $\varepsilon_0$  into  $\mathbb{Z}_2$ , defined by recursion on their Cantor normal forms.

The map (`cnfToZp2` in `Gentzen.lean`) sends the seed stage 0 to 0 itself and, from stage 1 on, the  $n$ -th tower stage to  $2^n$  in  $\mathbb{Z}_2$ . Read through the valuation, the stages climb:  $2^n$  has 2-adic valuation  $n$ , rising toward the infinite valuation of 0. Read through the norm, they fall: the 2-adic norm of  $2^n$  is  $(1/2)^n$ , which goes to 0 as  $n$  increases. So as the ordinal tower climbs toward  $\varepsilon_0$  from below, its 2-adic image converges to  $0 = \perp$  in the 2-adic metric, the point it started from. Two convergences, one tower.



Same tower, two perspectives. Teal (top): ordinal stages approach  $\varepsilon$ -zero. Indigo (bottom): their 2-adic norms approach 0.

Same tower stages, two perspectives. Teal (top row): ordinal stages  $1, \omega, \omega^\omega, \dots$  approach  $\varepsilon_0$  from below. Indigo (bottom row): the 2-adic norms of their images  $1/2, 1/4, 1/8, \dots$  converge to  $0 = \perp$ . ZP-L §VII: `snap_zp2_correspondence`.

This dual convergence is what `snap_zp2_correspondence` formalizes in ZP-L §VII: four jointly established facts about the same tower sequence — all stages below  $\varepsilon_0$  in ordinal order, all assigned  $c_0$  by the canonical map, their images converging to 0 in  $\mathbb{Z}_2$  (from stage 1 on the images are  $2^n$ , with 2-adic norm  $(1/2)^n$ ; `Gentzen.lean` §IV), and the canonical map snapping to  $c_1$  at  $\varepsilon_0$ .

## The Kleene Connection

ZP-K carries  $\perp$ 's computational face as a requirement of its Kleene structure: the class field `botCode_is_quine`, which asks the nominated code to satisfy `IsComputationalQuine` — a periodicity condition,  $\text{eval } c \ n = \text{eval } c \ (\text{encode } c + n)$  for every  $n$ , so the code's own Gödel number is a period of it. Kleene's second recursion theorem guarantees such a code exists. The condition does not by itself pin self-reference: the constant codes satisfy it, and they are what witnesses `infinite_quine_family` (ZP-K §VI). That the program and the bottom are one object is the framework's reading rather than a Lean equation — a program code and the lattice carrier are different types. ZP-L identifies  $\varepsilon_0$  as an ordinal fixed point: the first ordinal satisfying  $\omega^\alpha = \alpha$ , guaranteed to exist by the fixed-point theorem for normal functions. Both fixed-point proofs carry `Classical.choice` in their Lean formalizations; ZP-K Section IV tabulates the measured footprints. Whether these reflect the same underlying mathematical phenomenon — or are coincidentally parallel — is the open question `Gentzen.lean` §I poses.

ZP-L §VI (`kleene_ordinal_snap_bridge`) names this structural parallel explicitly. The theorem itself is purely ordinal — no `Code` or `eval` object appears in it. What the section establishes is that the hypothesis structure of the two proofs is the same: diagonalization and fixed-point existence.

### Two rooms, same pattern

Imagine two mathematicians working in separate rooms. One (ZP-K) is computing with programs and asking: what program runs itself? The other (ZP-L) is computing with ordinals and asking: what ordinal satisfies  $\omega^\alpha = \alpha$ ? Neither knows what the other is doing. When they compare notes, they find the same proof structure: the same diagonal argument, the same kind of fixed-point existence result.

ZPM relates the two through their 2-adic encodings; a bridge between the settings themselves is still open. The two rooms share a floor plan.

## Convergence with Proof Theory

ZP-L derives  $\epsilon_0$  as the snap threshold from ordinal fixed-point structure alone —  $\omega$ -tower iteration, the fixed-point property of  $\alpha \mapsto \omega^\alpha$ , and the map `cnfToZp2` from Cantor normal forms into  $\mathbb{Z}_2$ . Remark R-L.1 in the formal document notes a structural alignment with an independent result from proof theory.

Two results pin  $\epsilon_0$  as the proof-theoretic ordinal of Peano Arithmetic — the limit of how far PA's own proofs of transfinite induction reach — and they run in opposite directions. Gentzen reports, crediting Hilbert-Bernays, that transfinite induction up to any ordinal strictly below  $\epsilon_0$  is provable in PA: every ordinal short of the boundary is reached. And transfinite induction up to  $\epsilon_0$  itself is not provable in PA, which Gentzen proved directly in 1943: the boundary itself is not reached. A boundary needs both sides — the side that is reached and the side that is not — and neither result supplies both. A third result stands behind the pair: transfinite induction up to  $\epsilon_0$  is enough to prove PA consistent (1936). That is a fact about what  $\epsilon_0$  steps of induction prove rather than about what PA proves, so it does not on its own single out  $\epsilon_0$ ; it is also the indirect route to the unprovability half, by way of Gödel's theorem together with that 1936 result.

ZP-L and Gentzen arrive at  $\epsilon_0$  from entirely separate starting points. ZP-L starts from ordinal fixed-point structure:  $\epsilon_0$  is where  $\omega$ -tower self-iteration closes on itself. Gentzen starts from proof theory:  $\epsilon_0$  is where PA's provability tower runs out. Neither derivation references the other's domain. Both locate the same boundary.

The convergence is notable. ZP-L's fixed-point derivation and Gentzen's proof-theoretic analysis were conducted in separate domains with separate machinery. That both locate  $\epsilon_0$  is a structural observation, not an argument for either result. Both stand or fall on their own proofs.

ZP does not prove any part of Gödel's incompleteness theorems, and does not reprove Gentzen's ordinal analysis of PA. The connection is a structural observation: two independent formal derivations arrive at the same ordinal boundary. Two results pin  $\epsilon_0$ , because pinning a boundary is two-sided: it needs witnesses below the boundary and a ceiling at it, and a third result stands behind the pair. The Convergence section above states each of them, with its credit; the formal ZP-L states them in Remark R-L.1. ZP-L established  $\epsilon_0$  as the snap threshold from ordinal fixed-point structure. No formal equivalence between the two derivations is claimed.

## A Note on Classical.choice

Every theorem in ZP-L's own Lean file (`Gentzen.lean`) carries the axiom footprint `[propext, Classical.choice, Quot.sound]`, inherited from Lean 4 and Mathlib rather than a novel ZP-L commitment. Results ZP-L cites

from other layers are a separate question: ZP-K Section IV tabulates measured footprints on both sides.

What is notable is where Classical.choice appears, and where it does not. Across the layers ZP-L §I tabulates, the footprints are not uniform; ZP-K Section IV tabulates the measured footprints. Whether Classical.choice is structurally forced by the ZP framework — rather than merely inherited from Mathlib infrastructure — remains an open question.

#### Key Result — ZP-L: The Ordinal Snap Threshold

`epsilon_zero_snap_canonical` (Gentzen.lean §VII): The canonical map  $\phi(\alpha) = c_0$  if  $\alpha < \epsilon_0$ , else  $c_1$  satisfies all five conditions simultaneously — monotone, tower-aligned ( $c_0$  on every fundamental sequence stage), fixed-point-respecting ( $c_1$  on every fixed point of  $\omega^{\cdot}$ ), snapping at  $\epsilon_0$ , and  $\epsilon_0$  minimal — with no free hypotheses. `snap_zp2_correspondence` co-proves that the same tower witnesses both the ordinal approach to  $\epsilon_0$  and the 2-adic convergence to  $0 = \perp$ . All theorems proved without sorry. ZPM relates the ordinal and machine-phase encodings through the 2-adics.